THE NOVA RESEARCH CURRICULUM

Biology

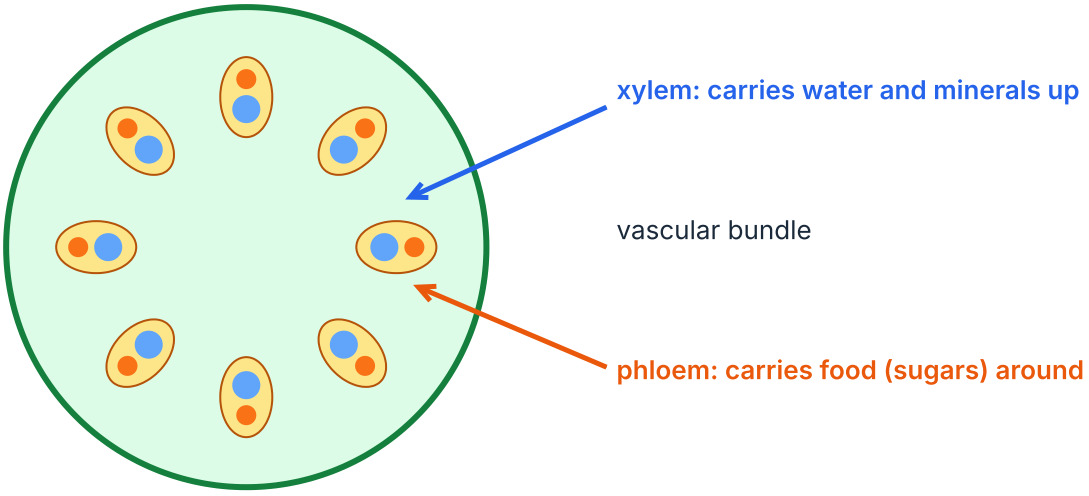
Grade 7 · Term 1

Student edition

Outcome: BI-7.1 plant transport, photosynthesis, respiration and transpiration.

Lesson notes

- **Plant transport (Weeks 1–4):** roots take in water and minerals through root hairs; xylem carries water up; phloem carries food (sugars) around the plant. Home lab: celery or a white flower in coloured water.



Cross-section of a stem: xylem (blue) and phloem (orange) in vascular bundles.

- **Photosynthesis (Weeks 5–8):** carbon dioxide + water → glucose + oxygen, using light energy and chlorophyll in the leaves. Leaf structure: stomata, guard cells. Teacher demonstration only: starch test on a leaf with iodine (heating alcohol is unsafe for students).

  ▶ **Try it on NovaSchool:** Photosynthesis in a leaf — open **Simulations** on your dashboard.

- **Respiration and transpiration (Weeks 9–12):** plants respire all the time: glucose + oxygen → carbon dioxide + water + energy. Transpiration: water evaporates from leaves through stomata and pulls water up the stem. Practical: plastic bag over a leafy branch; write a practical report (aim, method, results, conclusion).

Workbook pages

Page	Task
1	Label root, stem, xylem and phloem.
2–3	Photosynthesis word equation; label a leaf.
4–5	Transpiration practical report.

Question bank

#	Outcome	Question	Your answer
1	BI-7.1	Which tissue carries water up the stem?	phloem / xylem / stomata / chlorophyll
2	BI-7.1	Which gas do plants take in for photosynthesis?	oxygen / nitrogen / carbon dioxide / hydrogen
3	BI-7.1	The green pigment that traps light is …	glucose / chlorophyll / starch / xylem
4	BI-7.1	Water leaves the leaf through …	roots / stomata / phloem / flowers
5	BI-7.1	Plants respire only at night. True or false.	True / False
6	BI-7.1	Write the word equation for photosynthesis.	_____
7	BI-7.1	Iodine turns blue-black when … is present.	glucose / starch / water / oxygen
8	BI-7.1	Transpiration is faster on a hot, windy day. True or false.	True / False